

Cloud-Based Hospital Management and Analytics Dashboard Using Google Cloud Platform

Course- cloud computing {CDAC}

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Branch – BIOMEDICAL

Introduction

Hospitals generate large amounts of data every day, including patient records, doctor information, admission details, and billing information. Traditional methods of managing this data can be inefficient and difficult to analyze.

Cloud computing provides scalable and secure solutions for storing, processing, and visualizing healthcare data. This project demonstrates how Google Cloud Platform can be used to build a cloud-based hospital analytics system.

Objective -

Store hospital data on the cloud.

Analyze patient and hospital records.

Create interactive dashboards for data visualization.

Demonstrate cloud computing concepts using GCP services.

Provide real-time insights into hospital operations.

Software and Tools Used –

GCP , Looker studio , Github

Implementation Steps –

Step 1: Create Dataset

Step 2: Create GCP Project

Step 3: Create Cloud Storage Bucket

Step 4: Upload Dataset

Step 5: Create BigQuery Dataset

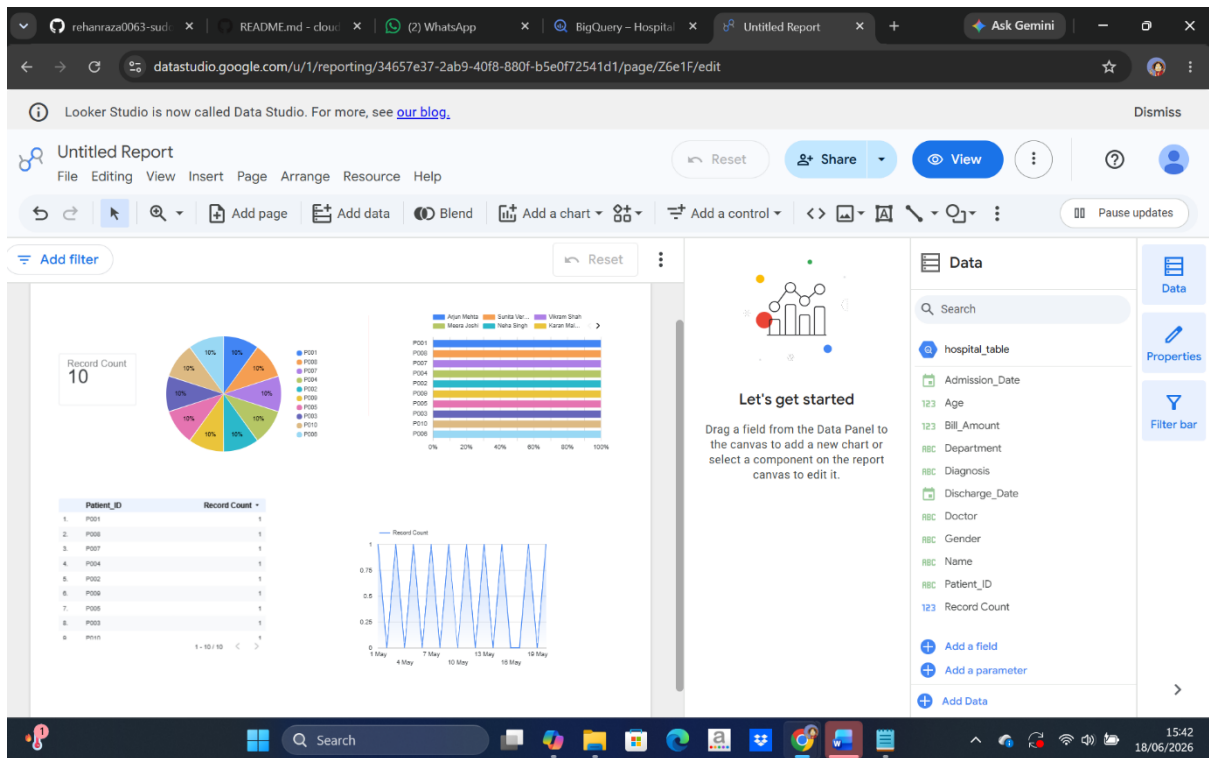
Step 6: Create BigQuery Table

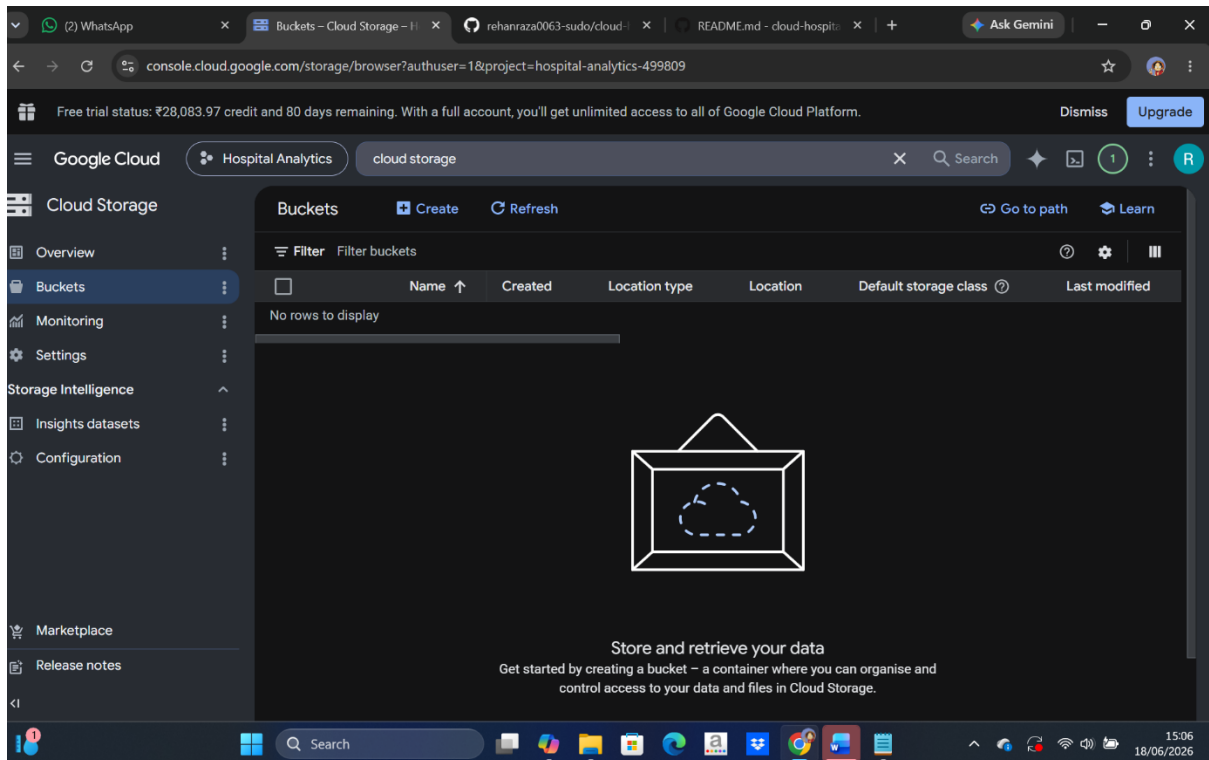
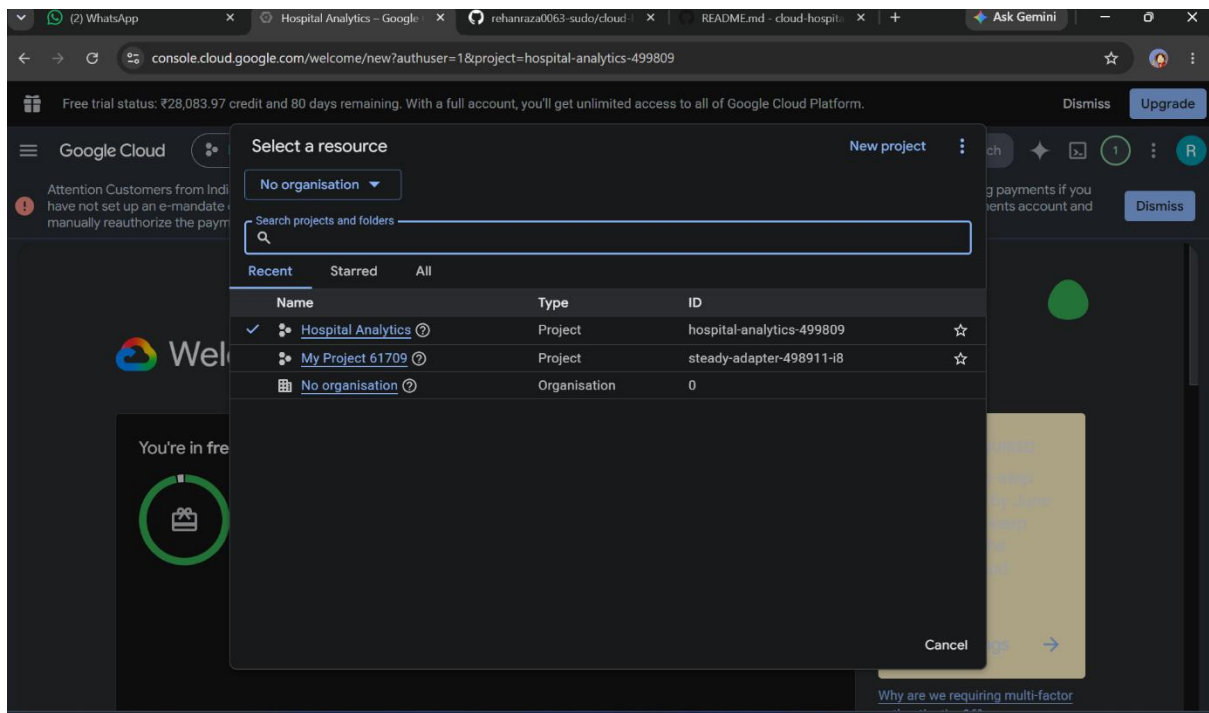
Step 7: Data Analysis Using SQL

Step 8: Create Dashboard in Looker Studio

Step 9: Create Visualizations

FINAL DASH BOARD IMAGE





Google CloudHospital Analyticscloud storage

Cloud Storage

OverviewBucketsMonitoringSettingsStorage IntelligenceInsights datasetsConfiguration

Create a Bucket

Get started

Pick a globally unique, permanent name. [Naming guidelines](#)

hospital-analytics-project

Tip: Don't include any sensitive information

Labels (optional)

Continue

Choose where to store your data

Location: us (multiple regions in the United States)

Good to know

Location pricing

Storage rates vary depending on data and location of your data

Current configuration: M

Item

us (multiple regions in the United States)

With default replication

Estimate your monthly

hospital-a...s-project - Cloud

rehanraza0063-sudo/cloud

README.md - cloud-hospita

(2) WhatsApp

Ask Gemini

console.cloud.google.com/storage/browser/hospital-analytics-project;tab=objects?authuser=1&project=hospital-analytics-499809&prefix=&forceOnObjectsS...

Free trial status: ₹28,083.97 credit and 80 days remaining. With a full account, you'll get unlimited access to all of Google Cloud Platform.

DismissUpgrade

Google CloudHospital Analyticscloud storage

Cloud Storage

OverviewBucketsMonitoringSettingsStorage IntelligenceInsights datasetsConfigurationMarketplaceRelease notes

Bucket details

hospital-analytics-project

Location: us (multiple regions in the United States)Storage class: StandardPublic access: Not publicProtection: Soft delete

ObjectsConfigurationPermissionsProtectionLifecycleObservabilityInventory ReportsOperations

Folder browser

hospital-analytics-project

Create folderUploadTransfer data

Filter by name prefix onlyFilter objects and foldersShowLive objects only

Name	Size	Type	Created	Storage class	Last modified
No rows to display					

15:09

18/06/2026

hospital-a-s-project - Cloud x rehanraza0063-sudo/cloud x README.md - cloud-hospit x (2) WhatsApp x Ask Gemini x

console.cloud.google.com/storage/browser/hospital-analytics-project;tab=objects?authuser=1&project=hospital-analytics-499809&prefix=&forceOnObjectsS...

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Google Cloud Hospital Analytics cloud storage

Cloud Storage

- Overview
- Buckets**
- Monitoring
- Settings
- Storage Intelligence
 - Insights datasets
 - Configuration
- Marketplace
- Release notes

Bucket details

hospital-analytics-project

Location: us (multiple regions in the United States) Storage class: Standard Public access: Not public Protection: Soft delete

Objects Configuration Permissions Protection Lifecycle Observability Inventory Reports Operations

Folder browser

hospital-analytics-project

Objects and folders

Create folder Upload Transfer data

Filter by name prefix only Filter objects and folders Show Live objects only

Name	Size	Type	Created
hospital_data.csv	952 B	text/csv	18 Jun 2

15:09 18/06/2026

BigQuery - Hospital Analytic x rehanraza0063-sudo/cloud x README.md - cloud-hospit x (2) WhatsApp x Ask Gemini x

console.cloud.google.com/bigquery?referrer=search&authuser=1&project=hospital-analytics-499809

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Google Cloud Hospital Analytics big query

BigQuery

- Overview Preview
- Studio**
- Agents Preview
- Pipelines and integration
 - Data transfers
 - Dataform
 - Scheduled queries
 - Scheduling
- Governance
 - Sharing (Analytics Hub)
- Partner Centre
- Settings Preview
- Release notes

Open project picker (Ctrl O)

Search BigQuery resources

Show starred only

hospital-analytics-499809

Create new

SQL query Notebook Notebook with Spark ML model

Graph Agent Conversation Pipeline Data preparation

Data canvas Table

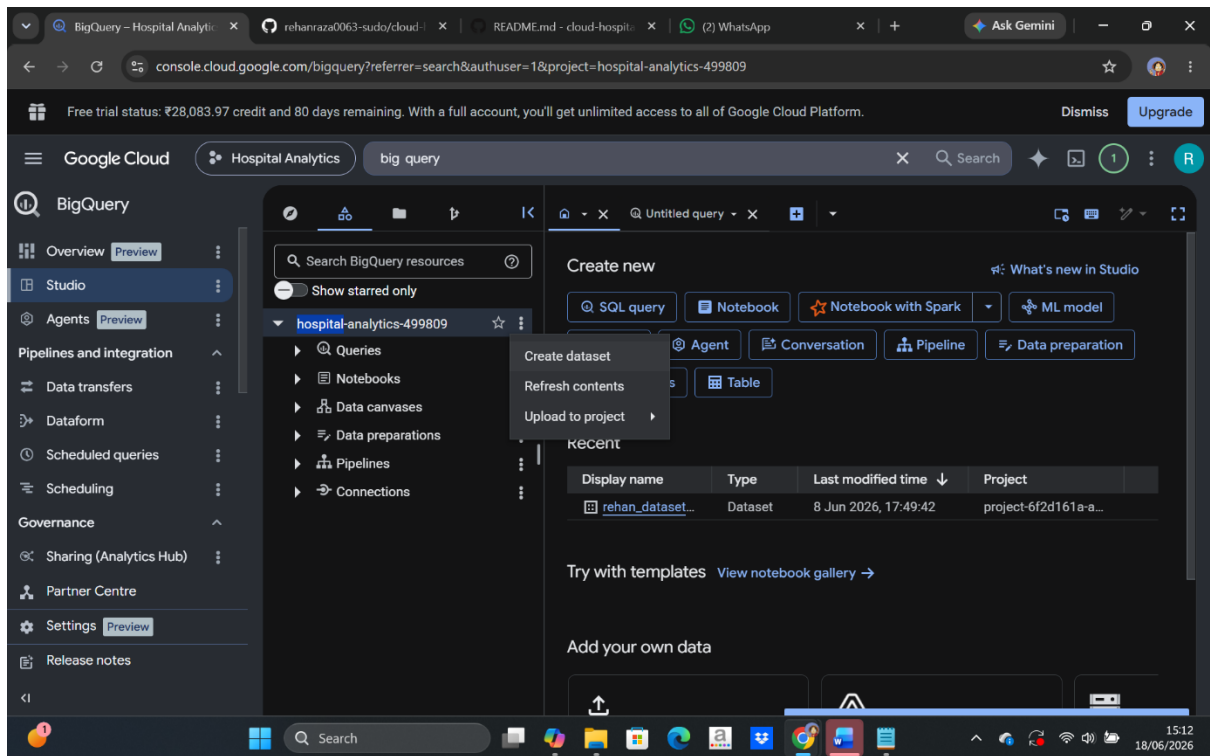
Recent

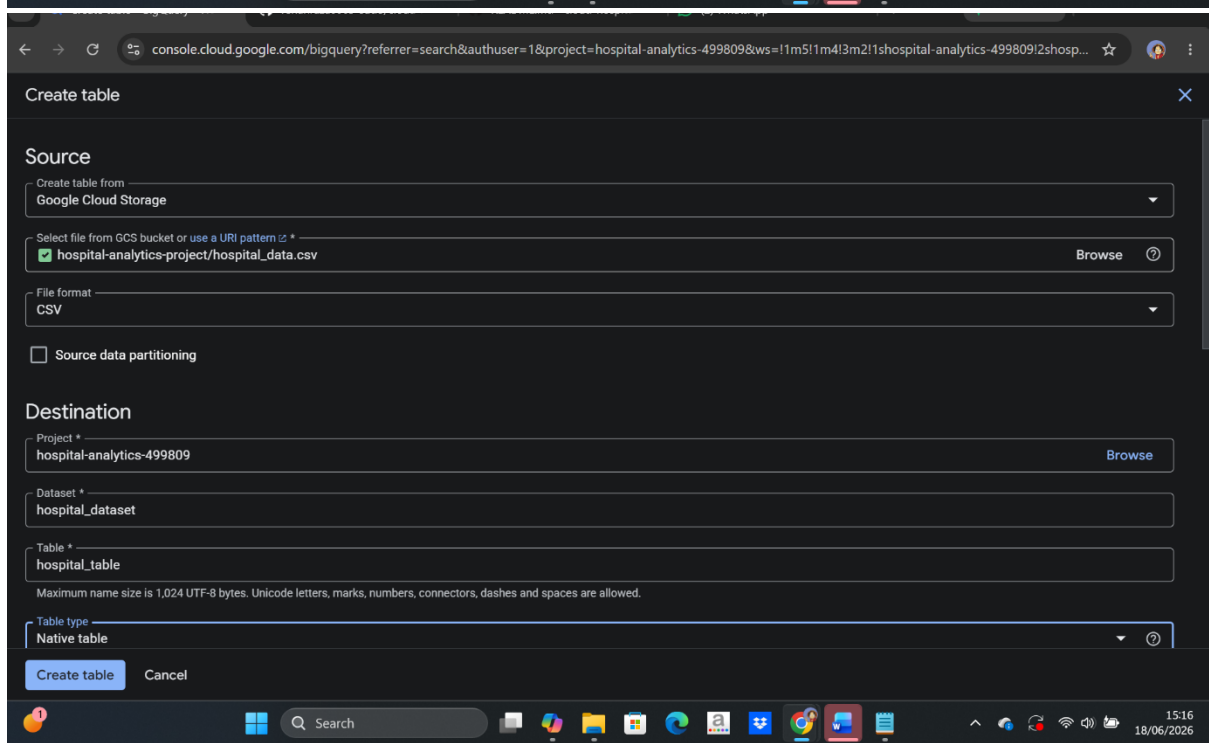
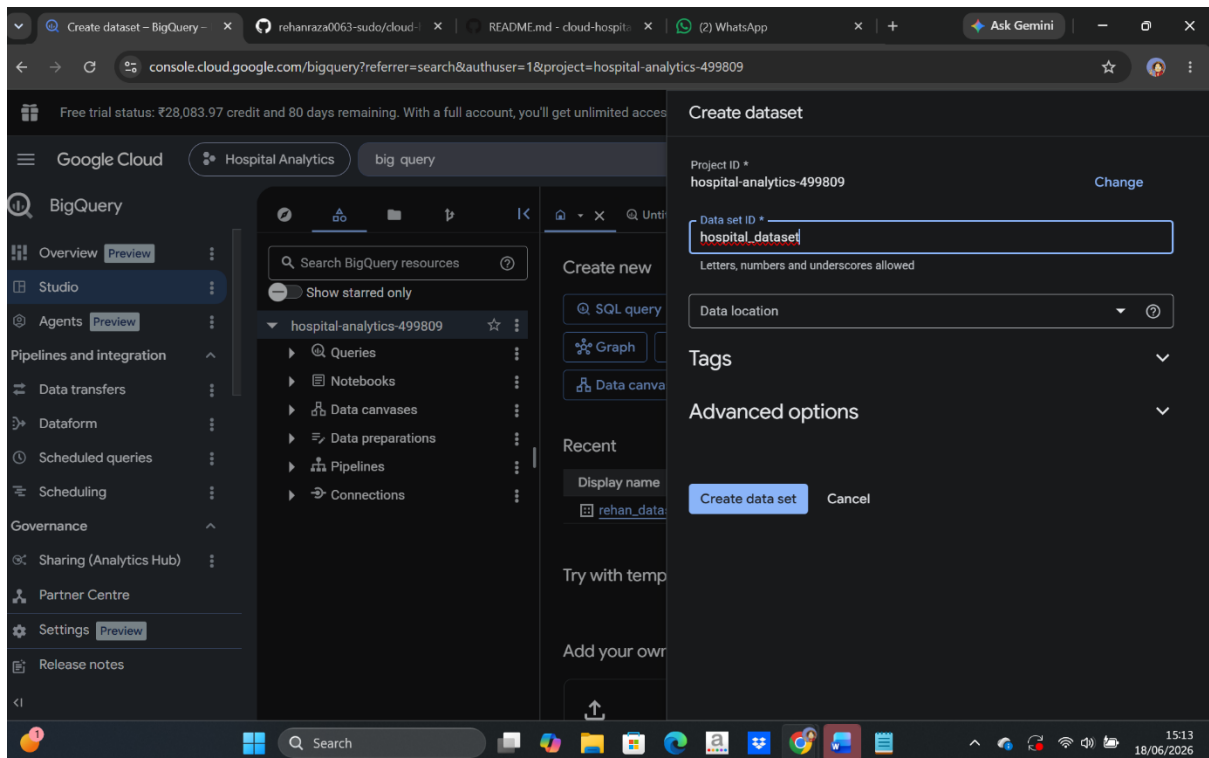
Display name	Type	Last modified time	Project
rehan_dataset...	Dataset	8 Jun 2026, 17:49:42	project-6f2d161a-a...

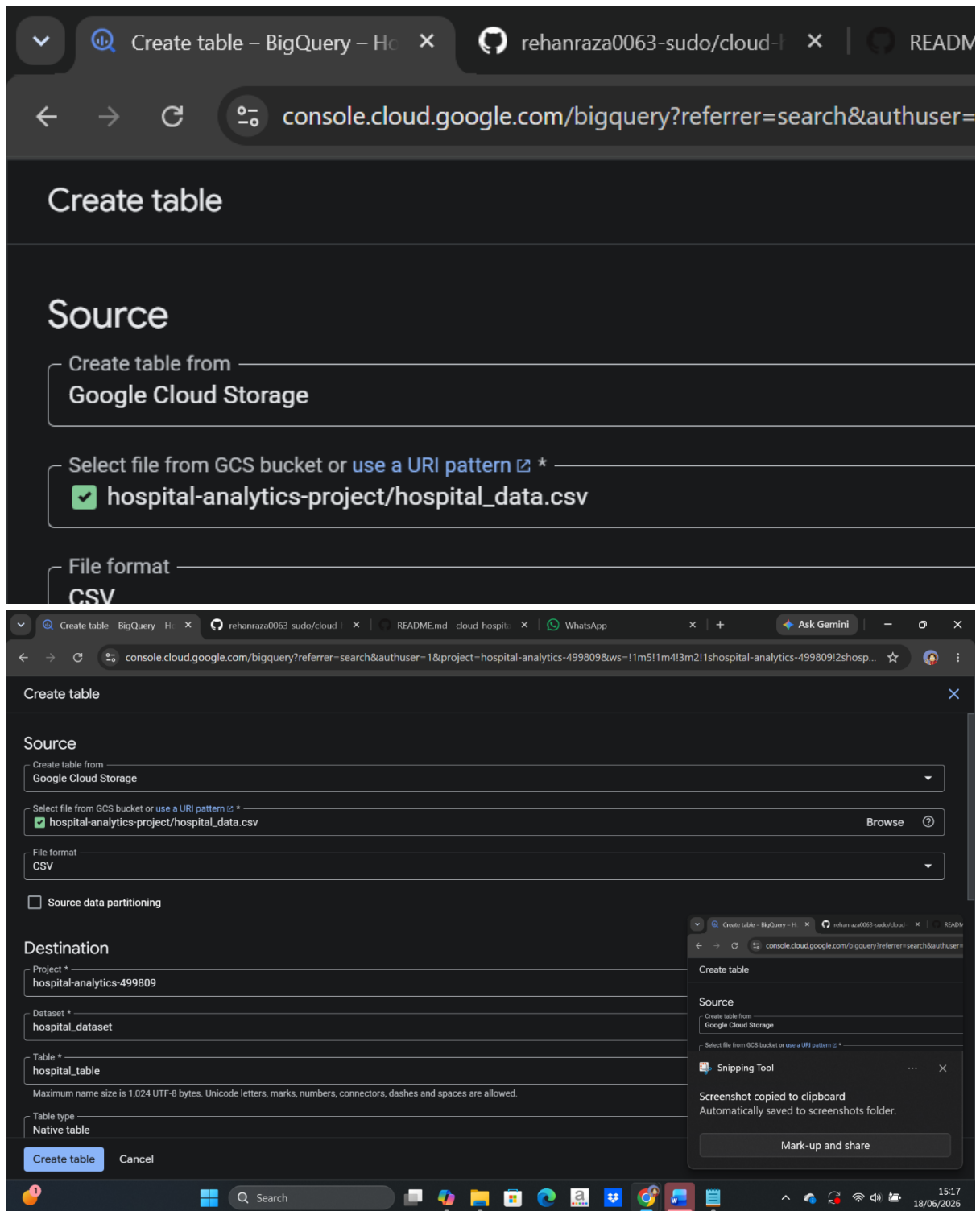
Try with templates View notebook gallery →

Add your own data

15:11 18/06/2026







Create table

Table *
hospital_table

Maximum name size is 1,024 UTF-8 bytes. Unicode letters, marks, numbers, connectors, dashes and spaces are allowed.

Table type
Native table

Schema

☒ Auto-detect

Schema will be automatically generated.

Partitioning settings

☒ No partitioning
☐ Partition by ingestion time
☐ Partition by field
To partition by field, you must first add fields to the schema above.

Cluster settings

Clustering order

Create table Cancel

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BigQuery

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Release notes

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- Queries
- Notebooks
- Data canvases
- Data preparations
- Pipelines
- Connections
- hospital_dataset
 - hospital_table

Show more

Untitled query

Run Save Schedule Share

File Edit Run Tools

```
1 SELECT COUNT(*) AS total_patients
2 FROM 'hospital_dataset.hospital_table'
```

This query will process 0 B when run.

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console.cloud.google.com/bigquery?referrer=search&authuser=1&project=hospital-analytics-499809&ws=11m611m511m311shospital-analytics-49980912sbqux...

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Show more

Untitled query Run Save Schedule Share

```
1 SELECT COUNT(*) AS total_patients
2 FROM `hospital_dataset.hospital_table`
```

Query completed

Query results Chat Save results Open in

Job information Results Visualisation JSON Execution details

Row	total_patients
1	10

Results per page: 50 1 - 1 of 1

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Search BigQuery resources

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- Queries
 - Shared queries
 - Untitled query 2026-06
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 - hospital_table

Show more

Untitled query Run Save Schedule Share

File Edit Run Tools

```
1 SELECT Department,
2 COUNT(*) AS patients
3 FROM `hospital_dataset.hospital_table`
4 GROUP BY Department
```

This query will process 119 B when run.

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 - Connections
- hospital_dataset
 - hospital_table

Show more

Untitled query Run Save Schedule Share

File Edit Run Tools

```
1 SELECT Department,
2 COUNT(*) AS patients
3 FROM `hospital_dataset.hospital_table`
```

Query completed

Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualisation JSON Execution details

Row	Department	patients
1	Cardiology	2
2	Dermatology	1
3	Gynecology	1
4	Neurology	2
5	Oncology	1
6	Orthopedics	2

Results per page: 50 1 - 7 of 7

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datastudio.google.com/u/1/navigation/reporting

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Data Studio

Search in Data Studio

Create

Recent

Shared with me

Owned by me

Bin

Templates

Conversational Analytics

Get started

Create a report

Chat with my data

Learn about Data Studio

Recent

Reports and data apps

Data sources

Explorer

Name	Type	Location	Owned by anyone	Last
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Create new report

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datastudio.google.com/u/1/reporting/34657e37-2ab9-40f8-880f-b5e0f72541d1/page/Z6e1f/edit

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Untitled Report

File View Page Help

Reset

Share

View

Pause updates

Add page

Add data

Blend

Add a chart

Add a control

Add data to report

Data credentials: Rehanrazagcp

BigQuery

By Google

BigQuery is Google's fully managed, petabyte scale, low-cost analytics data warehouse. BigQuery charges for querying/processing data. Those queries are charged to the credit card of the billing project.

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RECENT PROJECTS	Project	Data set	Table
MY PROJECTS	Search Projects	Search Datasets	Search Tables
SHARED PROJECTS			
CUSTOM QUERY	Hospital Analytics	hospital_dataset	hospital_table
PUBLIC DATA-SETS	My Project 61709		
	My First Project		

Cancel

Add

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Untitled Report
File Editing View Insert Page Arrange Resource Help

Reset Share View ?

↶ ↷ ↵ 🔍 + Add page + Add data 🔌 Blend 📊 Add a chart + Add a control < > 📄 📐 📏 📏 : ⏸ Pause updates

Add filter

Freeform layout Responsive layout

Let's get started
Drag a field from the Data Panel to the canvas to add a new chart or select a component on the report canvas to edit it.

Data

Search

hospital_table

Admission_Date

123 Age

123 Bill_Amount

ABC Department

ABC Diagnosis

Discharge_Date

ABC Doctor

ABC Gender

ABC Name

ABC Patient_ID

123 Record Count

+ Add a field

+ Add a parameter

+ Add Data

Data Properties Filter bar

Record Count
10

Patient_ID	Record Count
P001	1
P008	1
P007	1
P004	1
P002	1
P009	1
P006	1
P003	1
P010	1
mean	1

Admission Date	Discharge Date	Department	Diagnosis	Record Count
2023-05-01	2023-05-02	Internal Medicine	Myocardial Infarction	1
2023-05-01	2023-05-02	Internal Medicine	Myocardial Infarction	1
2023-05-01	2023-05-02	Internal Medicine	Myocardial Infarction	1
2023-05-01	2023-05-02	Internal Medicine	Myocardial Infarction	1
2023-05-01	2023-05-02	Internal Medicine	Myocardial Infarction	1
2023-05-01	2023-05-02	Internal Medicine	Myocardial Infarction	1
2023-05-01	2023-05-02	Internal Medicine	Myocardial Infarction	1
2023-05-01	2023-05-02	Internal Medicine	Myocardial Infarction	1
2023-05-01	2023-05-02	Internal Medicine	Myocardial Infarction	1
2023-05-01	2023-05-02	Internal Medicine	Myocardial Infarction	1

Record Count

1 May 4 May 7 May 10 May 13 May 16 May

Let's get started
Drag a field from the Data Panel to the canvas to add a new chart or select a component on the report canvas to edit it.

Data

Search

hospital_table

Admission_Date

123 Age

123 Bill_Amount

ABC Department

ABC Diagnosis

Discharge_Date

ABC Doctor

ABC Gender

ABC Name

ABC Patient_ID

123 Record Count

+ Add a field

+ Add a parameter

+ Add Data

Data Properties Filter bar

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